

PLUMBING & DRAINAGE BLUEPRINT

G+1 Farm Storage + Residence, Moradabad, UP

G+1 फार्म स्टोरेज + आवास - जल निकासी एवं नलसाजी

Building / भवन	20ft x 25ft internal, G+1 (Storage + Residence)
Systems / प्रणाली	4 SEPARATE drainage systems (Black + Grey + Wash + Rain)
Borewell / बोरवेल	30ft right of building, toward front (drinking water)
Road / चकरोड	10-12ft in front of building (kachchi road)
Septic Tank / सेप्टिक	5ft x 2.5ft x 4.5ft, 2-chamber RCC, behind building (left)
BW Soak Pit	5ft dia x 7ft deep, back-left (x=-5, y=40)
GW Soak Pit	4ft dia x 6ft deep, back-center (x=5, y=40)
WW Soak Pit	5ft dia x 6.5ft deep, back-right (x=15, y=38)
Min. spacing / दूरी	10ft between all pits (IS 2470)
Borewell safety	All pits 50ft+ from borewell
Pipe material	PVC SWR (IS 14735) - Ashirvad / Supreme / Finolex
Joints / जोड़	Solvent cement (underground). No rubber rings for buried pipes.
Concrete	M20 for septic tank (1:1.5:3)
Slope / ढलान	Black water: 1:40. Grey/Wash water: 1:60
Bends / मोड़	ONLY 45 degree Y-junctions. NO 90 degree bends for sewage.
Codes / कोड	IS 2470 (Septic), IS 1742 (Drainage), NBC 2016 Part 9
Date / दिनांक	June 2026
Drawing No.	PLB-001

COLOR CODE / रंग कोड:

	BLACK WATER (काला पानी - शौचालय) - Dark Maroon
	GREY WATER (धूसर पानी - रसोई/नहान) - Grey
	WASH WATER (धुलाई पानी - फर्श/दर्रेक्टर) - Ochre/Sandy
	RAINWATER (बारिश का पानी - छत) - Blue

BW PIT 5ft x 7ft

GW PIT 4ft x 6ft

WW PIT 5ft x 6.5ft

SEPTIC 5x2.5x4.5ft

OIL TRAP

SILT TRAP

GREASE TRAP

BUILDING 20x25ft

GF WC

1F WC

CHAKRODE / चक्रोडे (kachchi road)

SPACING / दूरी:

Septic to BW Pit: 12ft

BW Pit to GW Pit: 10ft

GW Pit to WW Pit: 10.2ft

All pits to borewell: 50ft+

Pits to building: 5ft+

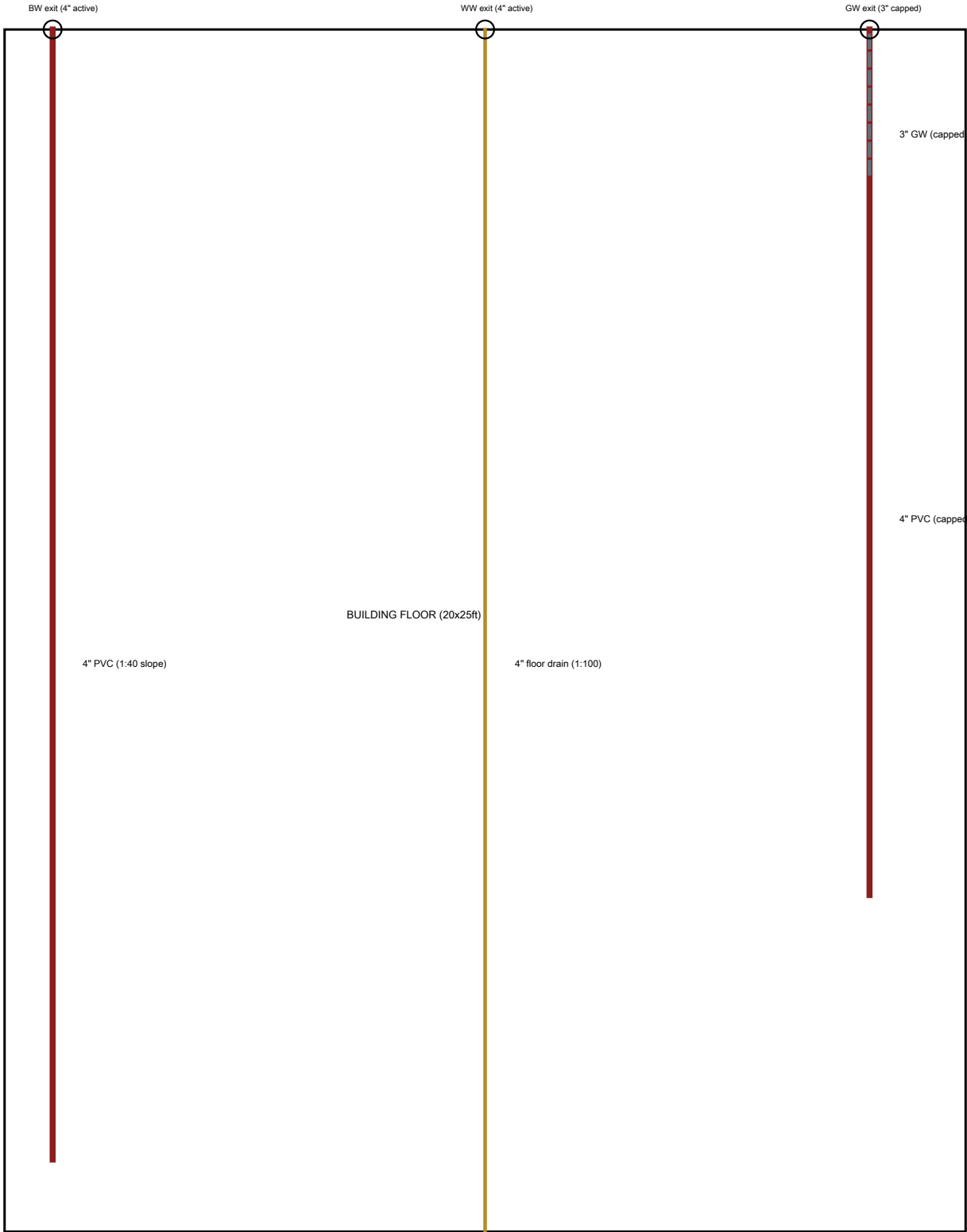
Left side: CLEAR (tractors)

BOREWELL / बोरे

50ft safety radius

SHEET 3: UNDERGROUND PIPE LAYOUT (Before Floor Slab Pour)

ALL pipes must be laid BEFORE GF floor slab is poured. Show pipe sizes, slopes, and exit points.



CRITICAL NOTES / ज़रूरी बातें:

- ALL 3 pipe routes MUST be laid BEFORE floor slab.
तीनों पाइप मार्ग फर्श की डलाई से पहले बछियाएँ!
- Pipe 1 (BW): Active immediately (toilet connected).
- Pipe 2 (GW): CAPPED both ends (open when 1F built).
- Pipe 3 (WW): Active (floor drain channel connected).
- 150mm sand bedding below and sides of all pipes.
पाइप के नीचे और बगल में 6 इंच रेत बछियाएँ!
- Backfill with selected earth (no stones/bricks).
चुनी हुई मट्टि से भराई (पत्थर/ब्रिक नही)।
- Minimum 1ft cover above pipes before slab.
- Check slope with spirit level before backfill.
भराई से पहले डलान स्पिरिट लेवल से जाँचें।

SHEET 4: SEPTIC TANK DETAIL / सेप्टिक टैंक विवरण

2-Chamber RCC tank. Position: behind building, left side (x=-5, y=28).

A. SEPTIC TANK SPECIFICATIONS / वशिपताएँ

Parameter / मद्	Value / मान	Notes / टिप्पणी
Size / आकार	5ft x 2.5ft x 4.5ft deep	External dimensions / बाहरी माप
Internal	4ft x 1.5ft x 4ft deep	After 6" walls / 6 इंच दीवार के बाद
Capacity / क्षमता	~1500 litres	For 3-5 users toilet only
Type / प्रकार	2-chamber RCC	Settling + Clarifying
Wall thickness	6" RCC (M20)	150mm, waterproof inside
Base slab	6" RCC (M20)	With 150mm PCC below
Top slab / ढक्कन	6" RCC with 2 manholes	18"x18" openings for cleaning
Chamber 1 (settling)	3.3ft long	2/3 of tank length
Chamber 2 (clarifying)	1.2ft long	1/3 of tank length
Baffle wall / बैफल	4" RCC, openings at 2/3 depth	Prevents short-circuiting
Inlet pipe	4" (110mm) PVC, 6" below top	With T-piece for scum retention
Outlet pipe	4" (110mm) PVC, 9" below top	With T-piece, to soak pit
Waterproofing	2 coats bitumen emulsion inside	Before backfilling
Cleaning / सफाई	Every 2-3 years (tanker pump)	सेप्टिक टैंकर से खाली करें
Position / स्थान	x=-5, y=28 (behind bldg, left)	Top flush with ground / ज़मीन बराबर

B. SOAK PIT SPECIFICATIONS / सोक पिट

Parameter	BW Pit (काला)	GW Pit (धूसर)	WW Pit (धुलाई)
Diameter / व्यास	5ft (1.5m)	4ft (1.2m)	5ft (1.5m)
Depth / गहराई	7ft (2.1m)	6ft (1.8m)	6.5ft (2.0m)
Position (x, y)	x=-5, y=40	x=5, y=40	x=15, y=38
Wall / दीवार	Honeycomb brick	Honeycomb brick	Honeycomb brick
Gravel bed	2ft bottom	1.5ft bottom	1.5ft bottom
Cover / ढक्कन	8" RCC (tractor rated)	6" RCC	6" RCC
Dist. from borewell	68ft	60ft	51ft

A. PIPE SCHEDULE / पाइप वविरण

Pipe Run / मार्ग	Size	Material	Slope	Bends	Notes / टपिपणी
GF toilet to back wall	4" (110mm)	PVC SWR	1:40	None	Active now / अभी चालू
1F toilet to back wall	4" (110mm)	PVC SWR	1:40	None	Active now / अभी चालू
Behind bldg (BW merge)	4" (110mm)	PVC SWR	1:40	45 deg Y	Y-junction, not 90 deg!
Y-jn to Septic	4" (110mm)	PVC SWR	1:40	None	Short run ~5ft
Septic to BW pit	4" (110mm)	PVC SWR	1:60	None	Gentle slope / हल्की ढलान
GW fixtures to downpipe	3" (75mm)	PVC SWR	1:40	45 max	Inside 1F floor
GW downpipe (vertical)	3" (75mm)	PVC SWR	Vertical	None	Outside back wall
GW to grease trap	3" (75mm)	PVC SWR	1:40	45 deg	Underground
Grease trap to GW pit	3" (75mm)	PVC SWR	1:60	None	Underground
Floor drain channel	6" wide	Cement	1:100	N/A	In GF floor slab / फर्श में
WW exit to silt trap	4" (110mm)	PVC SWR	1:60	None	Exits back wall
Silt to oil trap	4" (110mm)	PVC SWR	1:60	None	Underground
Oil trap to WW pit	4" (110mm)	PVC SWR	1:60	None	Underground
Vent pipe (toilet)	2" (50mm)	PVC	Vertical	None	GF toilet to above roof
Rainwater downpipes	3" (75mm)	PVC	Vertical	None	4 nos at corners

B. PIPE BRANDS (Moradabad) / उपलब्ध ब्रांड

- PVC SWR Pipes: Ashirvad (preferred) / Supreme / Finolex / Prince
- Solvent Cement: Ashirvad/Supreme brand (matching pipe brand)
- CI Covers (manholes): Heavy-duty Class D - local Moradabad foundry
- Floor Traps: Jaquar / Hindware / Parryware (CP brass)
- Nahani Trap: PVC with SS jali - 6"x6" for bathroom, 4"x4" for kitchen

SHEET 6: CONSTRUCTION SEQUENCE / निर्माण क्रम

Step-by-step for the mistri/contractor. ठेकेदार/मस्तिरी के लिए कदम-दर-कदम।

PLUMBING CONSTRUCTION SEQUENCE / नलसाजी कार्य क्रम

1. नींव खोदने के बाद (After foundation digging):
- a) सेप्टिक टैंक का गड्ढा खोदें (x=-5, y=28) - 6ft x 3.5ft x 5.5ft

b) 3 सोक पिट के गड्ढे खोदें (see Sheet 4 for positions)

c) सेप्टिक टैंक बनाएँ: PCC bed > base slab > walls > baffle > lid

d) सोक पिट बनाएँ: gravel bed > honeycomb brick > RCC cover
2. प्लिथ भरवाई से पहले (Before plinth fill):
- a) 3 pipe trenches खोदें (GF floor level से 1.5ft नीचे)

b) 150mm रेत बछियाँ (sand bedding)

c) Pipe 1 (BW): toilet position (x=1,y=1.5) to back wall exit

d) Pipe 2 (GW): stub at (x=18,y=22) to back wall - CAP BOTH ENDS

e) Pipe 3 (WW): from center (x=10) to back wall exit

f) Slope check with spirit level! ढलान जाँचें!

g) Backfill carefully with selected earth (मिट्टी से भरवाई)
3. प्लिथ भरवाई के बाद / GF फ्लोर से पहले (After plinth, before GF slab):
- a) Floor drain channel shuttering (x=10, 6" wide, full depth)

b) Toilet floor trap position mark (x=1, y=1.5)

c) All pipe stubs should be visible above fill level
4. GF फ्लोर ढलाई (GF floor slab pour):
- a) Pipe stubs sealed with plastic caps during pour

b) Floor drain channel cast-in with slope (1:100 toward back)

c) After curing: connect toilet to pipe stub
5. बाहर का काम (External work - can be done anytime):
- a) Connect BW exit pipe to inspection chamber to septic

b) Connect septic outlet to BW soak pit

c) Connect WW exit to silt trap > oil trap > WW pit

d) Lay GW underground route (capped) to grease trap location

e) Install rainwater downpipe sleeves through plinth wall
6. पहली मंजलि बनने के बाद (After 1F is built):
- a) Open GW pipe cap, connect to 1F kitchen/bath downpipe

b) Install grease trap at position (x=15, y=28)

c) Connect grease trap to inspection chamber to GW soak pit

d) Install 1F toilet, connect to BW vertical pipe (already in wall chase)

A. INSPECTION CHAMBERS / निरीक्षण चेम्बर

- Size: 1.5ft x 1.5ft x 2ft deep (450mm x 450mm x 600mm)
- Construction: Brick masonry with 1:4 cement mortar plaster inside
- Cover: CI (Cast Iron) heavy-duty Class D frame + cover (40 tonne rated)
- Benching: Smooth cement slope inside to direct flow (half-channel at bottom)
- Quantity: 2 nos (1 for BW system, 1 for GW system)
- Position: BW chamber at (x=-4, y=27.5). GW chamber at (x=12, y=33).
- Purpose: Access for cleaning blockages. Tanker pump connection for septic.

B. GREASE TRAP / ग्रीस ट्रैप (Grey Water)

- Size: 600mm x 450mm x 600mm deep (2ft x 1.5ft x 2ft)
- Position: x=15, y=28 (behind building, near kitchen downpipe)
- Construction: RCC / Masonry with inlet and outlet at different heights
- Inlet: 3" pipe enters 2" below top
- Outlet: 3" pipe exits 4" below top (grease floats, clean water exits lower)
- Baffle: Vertical plate between inlet and outlet (forces water under)
- Cleaning: Scoop floating grease every 2-4 weeks
- Brand: Can use prefab FRP grease trap (Plasto/Supreme brand)

C. SILT TRAP & OIL TRAP / सिल्ट एवं ऑयल ट्रैप (Wash Water)

- SILT TRAP: 750mm x 750mm x 750mm (2.5ft cube) at x=9, y=28
- Purpose: Settles mud, sand from tractor wash / ट्रैक्टर धोने का कीचड़
- Cleaning: After every heavy wash / हर भारी धुलाई के बाद साफ करें
- OIL & GREASE TRAP: 750mm x 600mm x 750mm at x=8, y=33
- Purpose: Removes diesel, engine oil from wash water
- Construction: 2-chamber with baffle wall
- Cleaning: Monthly - scoop floating oil / हर महीने तेल नकालें
- Both traps: Masonry construction, CI covers, at ground level

D. IMPORTANT RULES / ज़रूरी नियम

- NEVER connect rainwater to any soak pit or septic!
बारिश का पानी कभी भी सेप्टिक/सोक पिट में न डालें!
- NEVER use 90-degree bends for black water pipes!
काले पानी की पाइप में 90 डिग्री मोड़ कभी न लगाएँ!
- NEVER mix black water and grey water systems!
काला पानी और धूसर पानी की लाइन अलग रखें!
- All soak pits MUST be 50ft+ from borewell!
सोक पिट बोरवेल से 50 फुट दूर होना ज़रूरी!

SHEET 8: RAINWATER SYSTEM & WATER SUPPLY / वर्षा जल एवं पानी की आपूर्ति

A. RAINWATER DRAINAGE / बारिश का पानी

- Downpipes: 4 nos, 3" (75mm) PVC at building corners
- Exit: Through plinth wall at ground level (sleeves pre-installed)
- Disposal: Surface drainage directed AWAY from building into farm
- Alternative: Rainwater recharge pit (3ft dia x 10ft, gravel-filled)
- IMPORTANT: Rainwater is NEVER connected to septic/soak pit!
- महत्वपूर्ण: बारिश का पानी सेप्टिक/सोक पिट से जोड़ना मना है!
- Roof drainage: Slab slope 1:100 toward corners (slope screed)
- Gutter: Not required (flat roof with parapet, water to downpipes)

B. WATER SUPPLY (FUTURE) / पानी की आपूर्ति

- Source: Borewell with submersible pump (30ft right of building)
- Tank: 1000L plastic tank on roof (2F level, on MS stand)
- Supply pipe: 1" (25mm) GI or CPVC from pump to tank
- Distribution: 3/4" (20mm) CPVC to kitchen, bathroom, toilet flush
- WATER SUPPLY TO BE DESIGNED SEPARATELY.
- This blueprint covers DRAINAGE only.
- Local brands for supply pipes:
 - CPVC: Ashirvad FlowGuard / Supreme CPVC / Astral
 - GI: Tata/Jindal (ISI marked)
 - Tank: Sintex / Supreme / Penguin (1000L, 3-layer)